School of Computer Science Engineering and Technology

Course- **BTech**
Course Code- **CSET340**

Year- **2026**
Date- **09-03-2026 to 13-03-2026**

Type- **Specialization Elective**
Course Name- **Advanced computer vision and video analytics**
Semester- **Even**
Batch- **2023-2027 (VI Sem)**

Lab Assignment No. 7

Exp. No.	Name	CO-1	CO-2	CO-3
1.	To perform various Image Enhancements techniques in Frequency domain	✓	✓	✓

Objective: Apply various image enhancements using different high pass and low pass filters on different images via the use of Python using OpenCV/Pillow/Ski Images Library? Perform the provided task as follows.

Data Set: Download the Zip_Folder containing "Test_Images_lab_7" images with some colored, gray scaled and black and white images.

1. Use the **Cat.jpg** image and convert it to its corresponding frequency domain magnitude using fast Fourier transform (FFT) and output its image. Now crop the low frequency and high frequency components from the magnitude spectrum and produce their output images after apply Inverse fast Fourier transform (IFFT)

   2 Marks

2. Use the grayscale image & apply the following image enhancement using low-pass filters (LPF) in the frequency domain to the image **(Penguine.jpg)**. Also display the **original image and filtered image** obtained after applying each filter.
   a. Ideal Low-Pass Filter (ILPF)
   b. Gaussian Low-Pass Filter (GLPF)
   c. Butterworth Low-Pass Filter (BLPF) **3 Marks**
3. Use the grayscale image & apply the following image enhancement using high-pass filters (HPF) in the frequency domain to the image **(CameraMan.jpg)**. Also display the **original image and filtered image** obtained after applying each filter.
   a. Ideal High-Pass Filter (IHPF)
   b. Gaussian High-Pass Filter (GHPF)
   c. Butterworth High-Pass Filter (BHPF) **3 Marks**
4. Use the **Cameraman.jpg** image and apply **gaussian smoothening** on it. Now, apply **homomorphic filtering** on it to enhance the image and show output image. **2 Marks**

Note:- **Suggested Platform:** Python: Jupyter Notebook/Visual Studio Code/Google Colab.
Mode of Delivery: Face-to-face: Instructor-led discussion and live coding demonstration. Hands-on Practice via Google Colab/VS code/Notebook.
Submission: On LMS within the prescribed time frame, else marks will.